



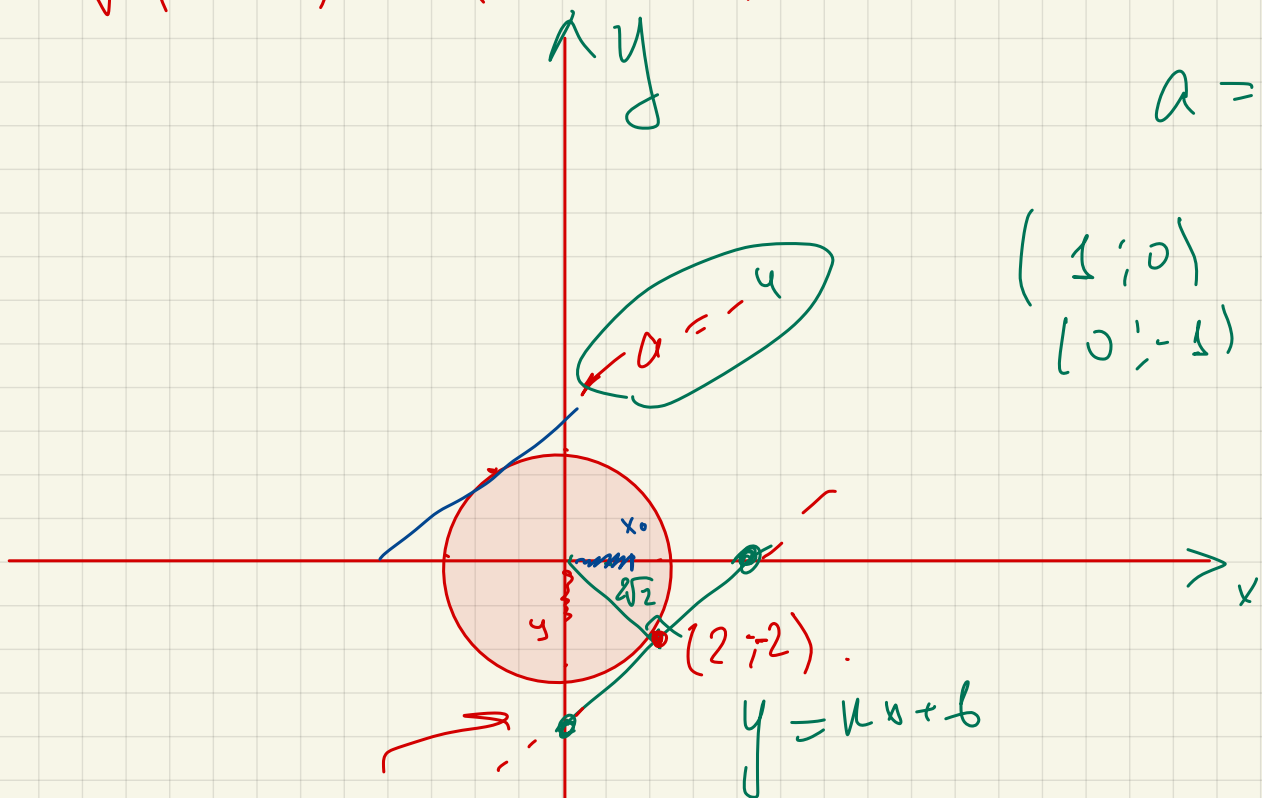
$$\sqrt{(x-a)^2 + y^2} + \sqrt{x^2 + (y+a)^2} = \underbrace{2a\sqrt{2}}$$

$(a; 0)$
 $(0; -a)$

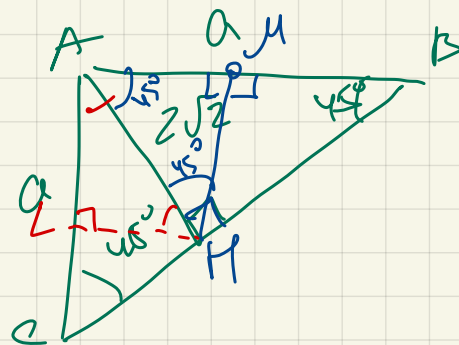
$$\sqrt{(a-0)^2 + (0-(-a))^2} = \sqrt{a^2 + a^2} = \underline{\underline{a\sqrt{2}}}$$

$$a = 1$$

$$\begin{pmatrix} 1; 0 \\ 0; -1 \end{pmatrix} \sqrt{2}$$



①



$$\angle A = 45^\circ$$

AB

$$AM = AB \cdot \sin 45^\circ$$

$$AM = AH \cdot \cos 45^\circ = 2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 2$$

$$AL = AM \cdot \cos 45^\circ = 2\sqrt{2} \cdot \frac{\sqrt{2}}{2} = 2$$

$$\begin{pmatrix} a:0 \\ 0:-a \end{pmatrix}$$

$$-a = b$$

$$0 = -ax + b$$

$$-ax = -a$$

$$x = 1$$

$$y = x - a$$

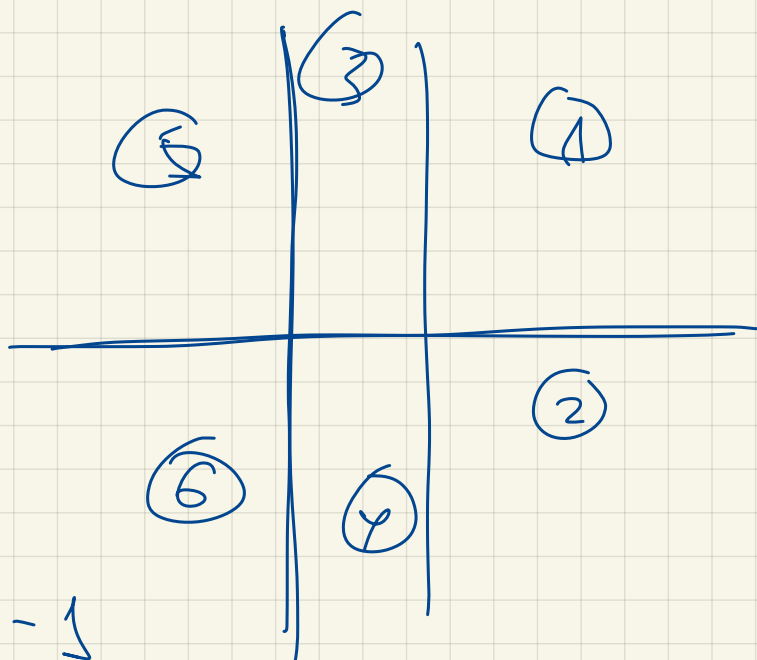
$$-2 = 2 - a$$

$$a = 4$$

$$\begin{cases} |y| + |2x - x^2| = 4 \\ y^2 + (2x - x^2)^2 = 4^2 \end{cases}$$

$$\begin{aligned} y &\geq 0 \\ y &< 0 \end{aligned}$$

$$2x - x^2 \xrightarrow{2} 0$$



$$-1 + 2x - x^2 = t - 1$$

$$0 \geq -(x-1)^2 = t - 1 \leq 0$$

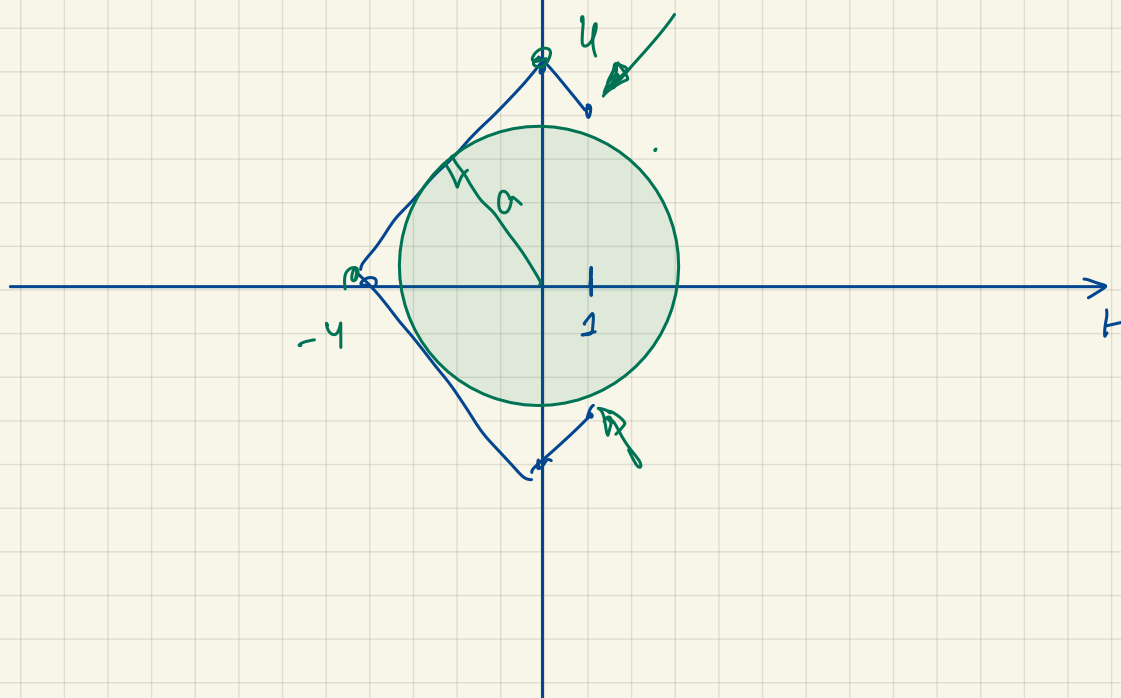
$$t \leq 1$$

$$\begin{cases} |y| + |t| = 4 \\ y^2 + t^2 = a^2 \\ t \leq 1 \end{cases}$$

$$-x^2 + 2x = t$$

$$R = |a|$$

$$a \in (-\sqrt{10}; -2\sqrt{2}) \cup (2\sqrt{2}; \sqrt{10})$$

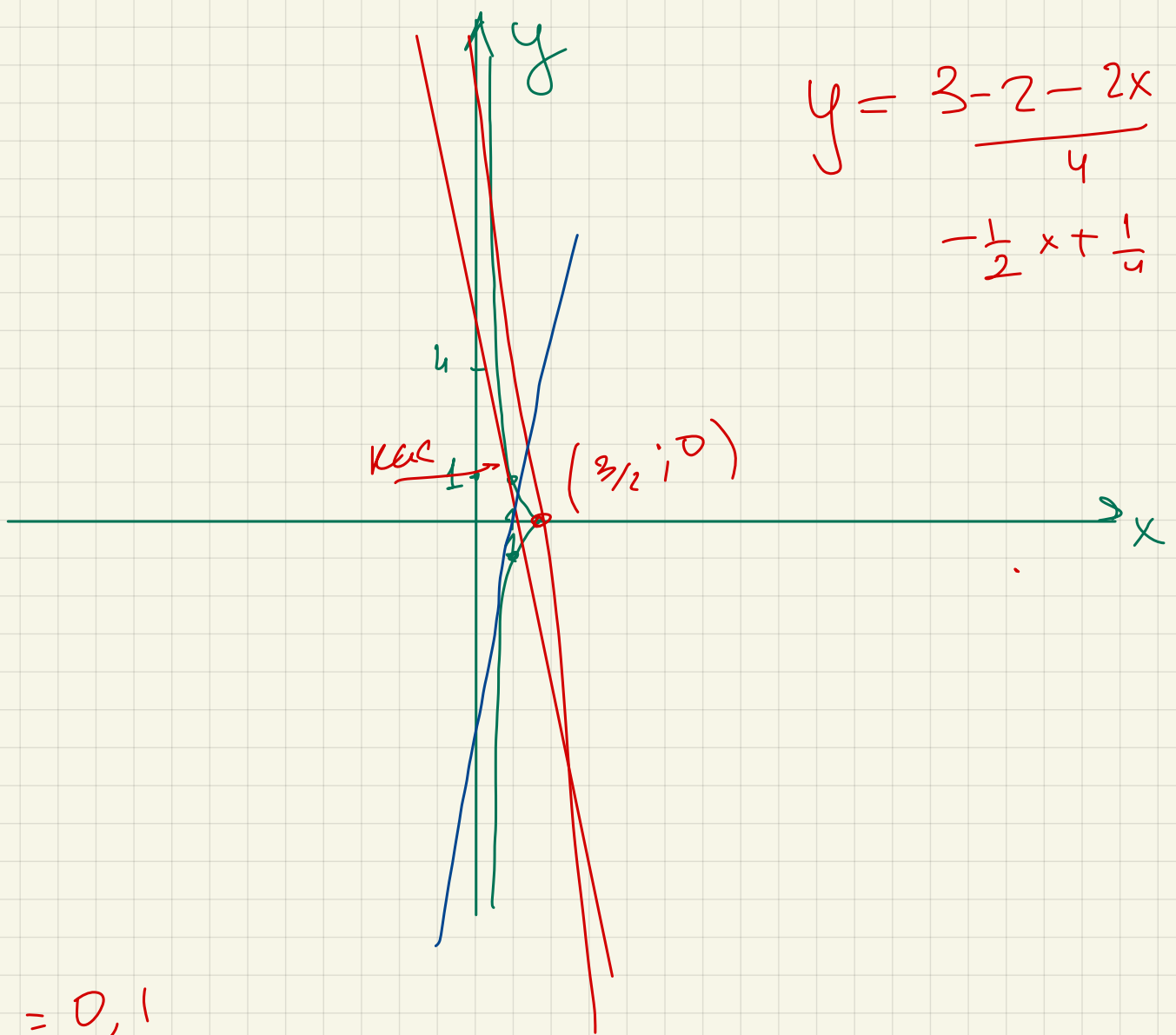


$$\begin{cases} 2x + 2ay + a - 3 = 0 \\ x|y| + 2x - 3 = 0 \end{cases} \quad \begin{cases} y = \frac{3-a-2x}{2a} \\ -\frac{2x}{2a} + \frac{3-a}{2a} \\ -\frac{x}{a} + \frac{3-a}{2a} \end{cases}$$

$$\begin{cases} y \geq 0 \\ y < 0 \end{cases} \quad \begin{cases} y = \frac{3-2x}{x} = \frac{3}{x} - 2 \\ y = \frac{2x-3}{x} = 2 - \frac{3}{x} \end{cases} \quad \begin{cases} a > 0 \end{cases}$$

$$\begin{aligned} a &= 1 \\ a &= 2 \end{aligned}$$

$$y = \frac{-2x+2}{2} = -x+1$$



$$y = \frac{3-2-2x}{4} = -\frac{1}{2}x + \frac{1}{4}$$

$$a = 0,1$$

$$y = \frac{3-2x-0,1}{0,2} = \frac{-2x}{0,2} + \frac{3-0,1}{0,2} = -10x + 15$$

$$y = \frac{3}{x} - 2 = \frac{3-2x-a}{2a}$$

$$\frac{3-2x}{x} = \frac{3-2x-a}{2a}$$

$$6a - 4ax = 3x - 2x^2 - ax$$

$$2x^2 - 3x - 3ax + 6a = 0$$

$$2x^2 - 3x(1+a) + 6a = 0$$

$$\Delta = 9(1+a)^2 - 24 \cdot 6a = 0$$

$$9(1+a^2+2a) - 48a = 0$$

$$3(1+a^2+2a) - 16a = 0$$

$$3a^2 - 10a + 3 = 0$$

$$\frac{10 \pm \sqrt{100 - 36}}{6} = \frac{10 \pm 8}{6} \rightarrow 3 \text{ or } \frac{1}{3}$$

$$a = \frac{1}{3}$$

$$0 = \frac{3 - 2 \cdot \frac{1}{3} - a}{2a} \rightarrow \phi$$

$$a \in \left(0, \frac{1}{3}\right] \rightarrow 2 \text{ per}$$

$$25a^2 - 78a + 9$$

$$\left(-; 0\right) \cup \left(0; \frac{1}{3}\right]$$